Milestone 2: Identifying Maternal Health Access and Risk Patterns Across U.S. Counties – White Paper

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DSC680 - Applied Data Science

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August 23, 2026

Technical Summary

- The final analytic file contains 578 named counties across all 50 states and 14.4 million births reported during 2020-2024.
- Low birth weight averaged 8.44% across counties. The median was 8.24%, and the observed county range was 5.21% to 15.77%.
- A random forest explained 57.1% of out-of-sample county variation. Its mean absolute error was 0.78 percentage points.
- Maternal age composition, poverty, population density, OB-GYN supply, and delayed or absent prenatal care were the strongest predictive signals in the fitted model.
- The findings can support screening and planning. They do not establish that changing one county-level factor will cause a specific improvement in birth outcomes.

Business Problem

Maternal and infant health conditions vary across U.S. counties, while outreach capacity and clinical resources also differ. Health systems, public health agencies, and community partners need a transparent method for identifying counties where adverse birth indicators and access constraints overlap. This project asks which measured county characteristics are most associated with low birth weight and how those patterns can support targeted investigation, outreach, and resource planning.

Background and History

Low birth weight is defined here as an infant weight below 2,500 grams. Birth weight is recorded on birth certificates and is available through the National Vital Statistics System. CDC WONDER provides aggregated natality counts by maternal residence and selected demographic, health, and medical characteristics. HRSA's Area Health Resources Files consolidate county-level workforce, facility, population, economic, and resource-scarcity measures from federal and other established sources. Prior research documents persistent maternal health disparities and cautions that differences reflect health care quality, access, clinical risk, and broader structural conditions (Howell, 2018).

Data Explanation and Preparation

Four CDC WONDER extracts were created for 2020-2024 by county of maternal residence: infant birth weight, trimester prenatal care began, tobacco use, and maternal age. Counts were pooled across five years to stabilize rates and reduce the effect of small cells. Records labeled Unidentified Counties were removed because they combine several counties with populations below CDC's county-identification threshold and cannot be linked to a single HRSA record.

The CDC extracts were joined to the 2024-2025 AHRF release with five-character county FIPS codes. The dependent variable is the percentage of births below 2,500 grams among births with known weight. Explanatory variables include delayed or absent prenatal care, maternal tobacco use, maternal age composition, poverty, insurance coverage, provider supply, obstetric-capable hospitals, population density, and primary care shortage status. Provider and hospital counts were converted to rates per 100,000 residents. Population density was log transformed because its distribution was highly right-skewed. Median imputation occurred inside each modeling pipeline to prevent information leakage during cross-validation.

Methods

Exploratory analysis summarized distributions, county rankings, bivariate relationships, and missingness. Two predictive models were evaluated with the same 10-fold shuffled cross-validation splits: ridge regression and random forest regression. Ridge regression provides a regularized linear benchmark. The random forest captures nonlinear patterns and interactions without requiring a prespecified functional form. Model quality was measured with out-of-sample R-squared and mean absolute error. Permutation importance measured the decrease in fitted R-squared after each feature was randomly shuffled. Importance reflects predictive contribution within this dataset, not causal effect.

Analysis

Low birth weight prevalence showed meaningful geographic variation. The middle half of counties ranged from 7.32% to 9.23%. The full observed range was wider, from 5.21% to 15.77%. Because the analysis is limited to named counties in CDC WONDER, these values describe larger reported counties and should not be generalized to every U.S. county.

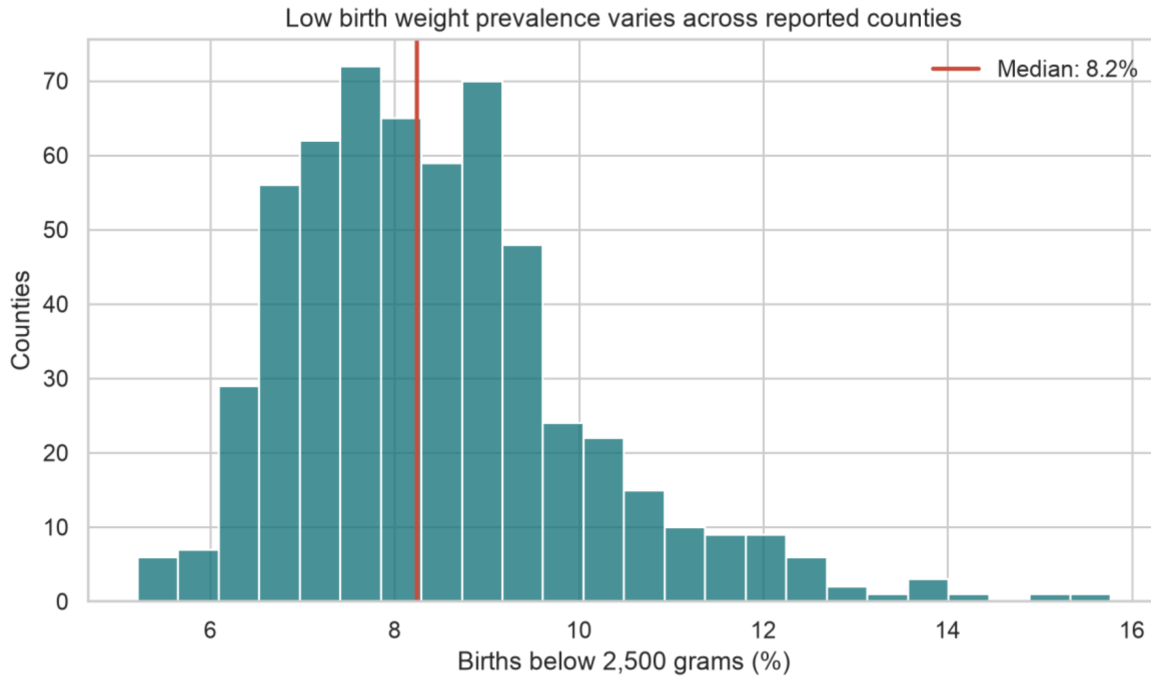

Figure 1. Distribution of pooled county low birth weight prevalence, 2020-2024. Source: CDC WONDER Natality.

The highest and lowest reported county values demonstrate that similar population thresholds do not produce uniform outcomes. Rankings are useful for screening, but they should be reviewed with count volume, uncertainty, reporting quality, and local context before action.

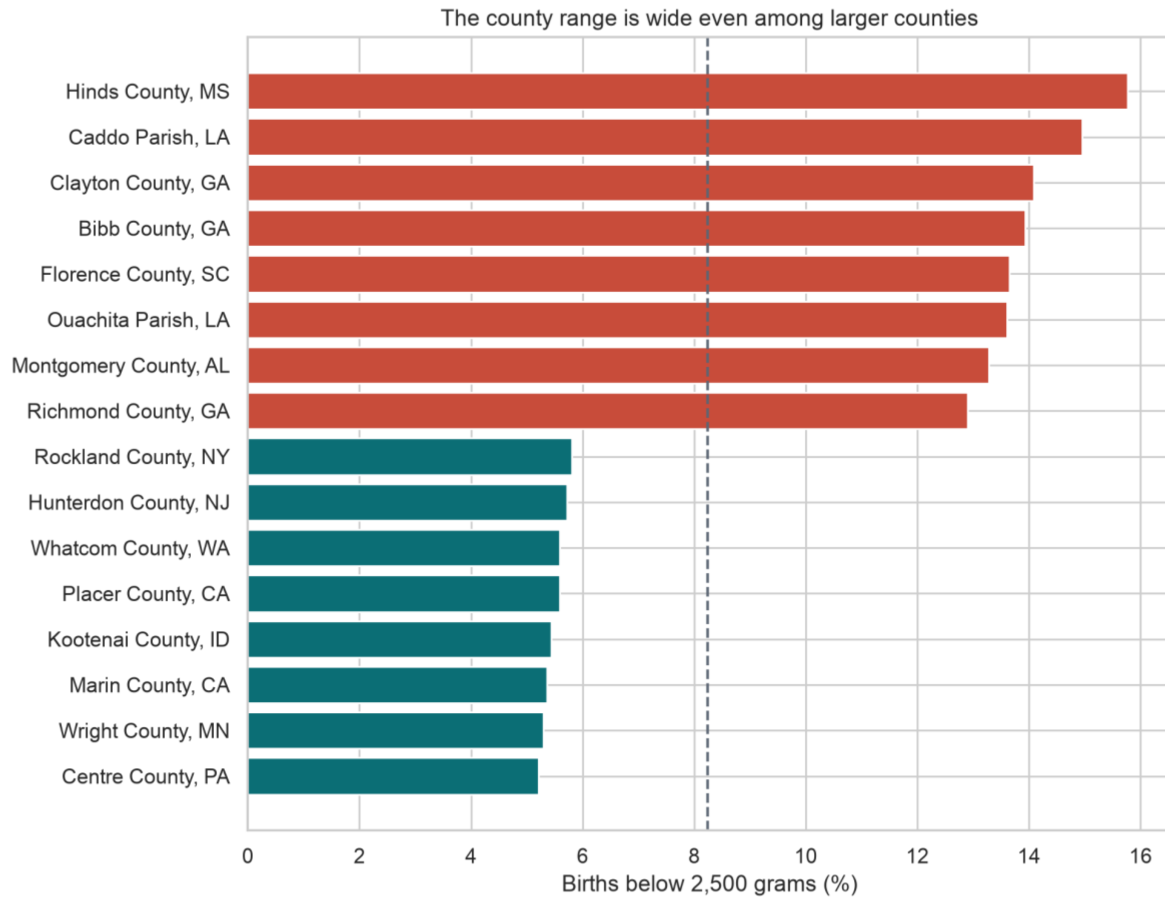

Figure 2. Counties with the eight highest and eight lowest observed rates among the analytic sample. Source: CDC WONDER Natality.

Poverty had a clear positive bivariate relationship with low birth weight, although counties at similar poverty levels still differed. That spread indicates that poverty alone does not explain the outcome and supports a multivariable approach.

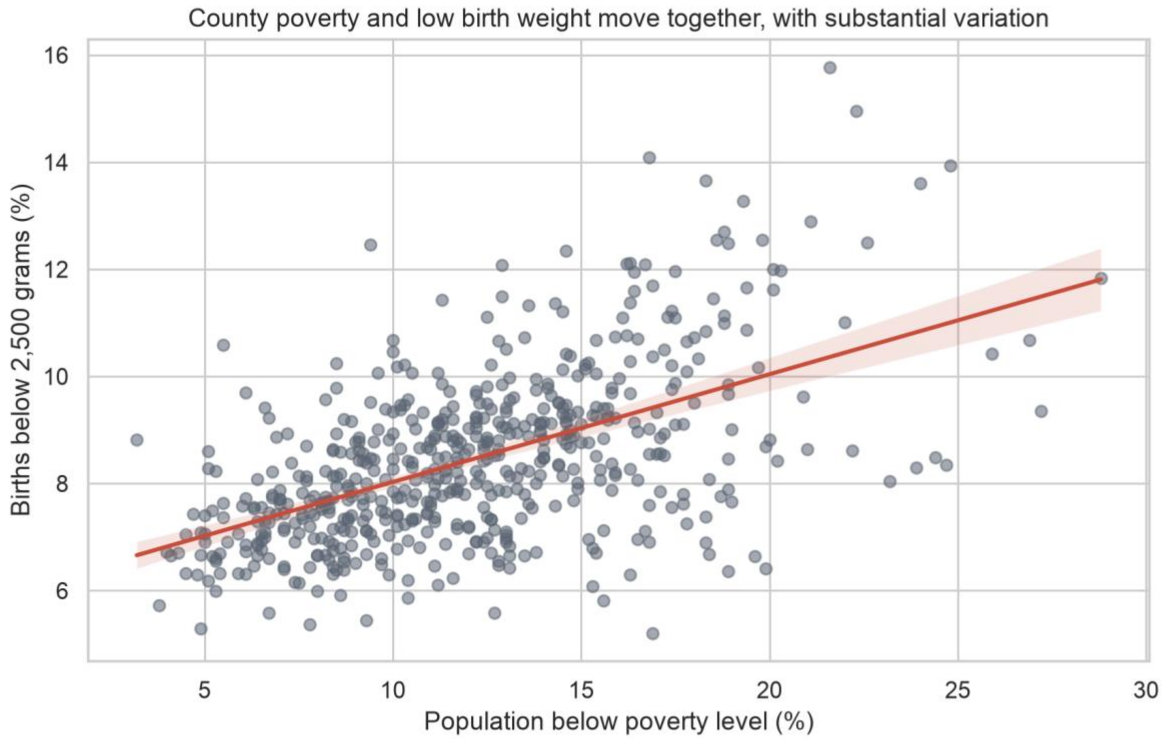

Figure 3. County poverty and low birth weight prevalence. Sources: CDC WONDER Natality and HRSA AHRF.

The random forest performed better than ridge regression in cross-validation, with an R-squared of 0.571 and a mean absolute error of 0.78 percentage points. The five highest permutation-importance features were maternal age 15 19 pct, poverty pct, log population density, obgyn per 100k, and late or no prenatal pct. Feature rankings reflect the information available to the fitted model. Correlated variables can share or shift importance, and provider supply may also proxy for urban concentration or referral patterns.

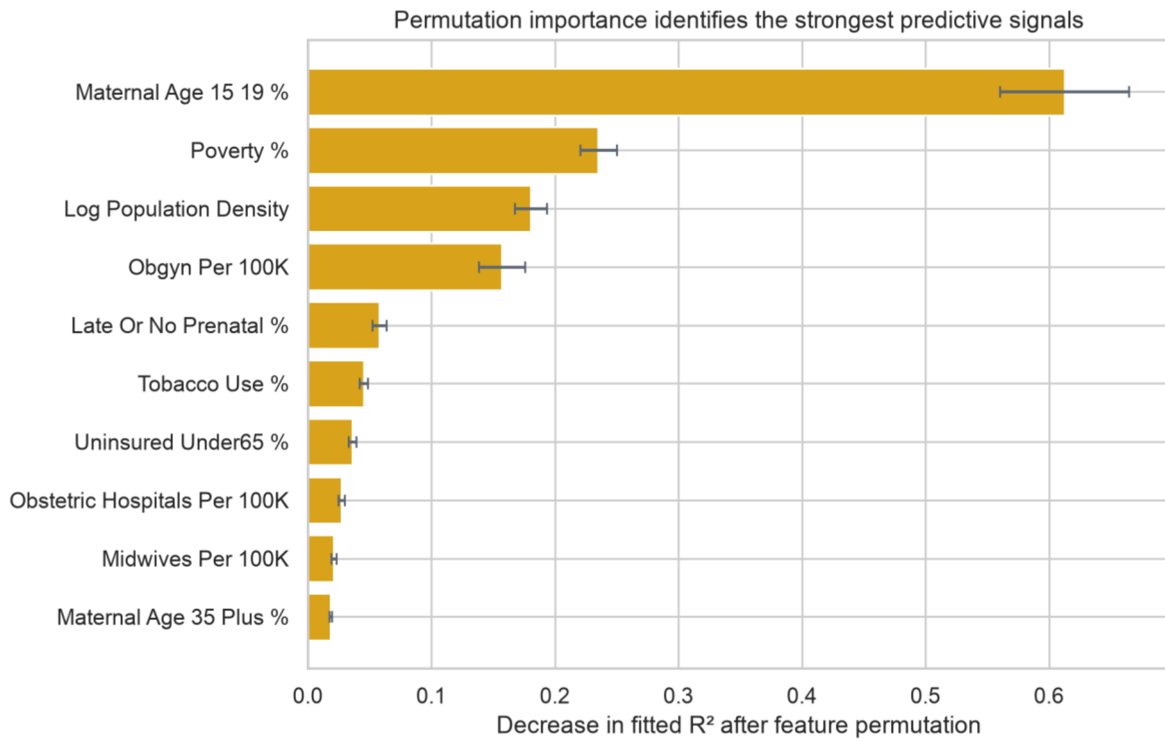

Figure 4. Random forest permutation importance using the full analytic sample after cross-validation comparison. Sources: CDC WONDER Natality and HRSA AHRF.

Conclusion

County-level low birth weight patterns were associated with both social conditions and health-access measures. Maternal age composition and poverty carried strong predictive information, while delayed prenatal care and provider supply added further signal. The model produced useful screening accuracy, while substantial unexplained variation remained. A defensible use is to identify counties for deeper review and local validation.

Assumptions

- County FIPS codes represent comparable geographic units across the linked files.

- Pooling 2020-2024 produces a stable recent-period outcome suitable for comparison with AHRF measures dated mainly 2022-2025.
- Births with known values are an adequate denominator for each percentage, and missing or suppressed cells do not change county rankings materially.
- County-level associations are relevant for planning even though they cannot describe individual risk.

Limitations

- CDC identifies counties only when the population meets its disclosure threshold. Smaller counties are grouped within each state, which excludes many rural counties from the linked analysis.
- Subnational counts from 1 through 9 are suppressed. Five-year pooling reduces suppression but does not remove it completely.
- The analysis is ecological and observational. It cannot establish causation or infer an individual's risk from county averages.
- Source years differ. The birth outcome spans 2020-2024, while AHRF measures use the most recent available year for each source.
- Provider counts do not measure appointment availability, travel time, service quality, insurance acceptance, or cross-county care use.
- Cross-validation estimates predictive performance within the observed sample. External validity for small or unidentified counties remains unknown.

Challenges

The main technical challenge was reconciling disclosure-limited natality geography with a complete county resource file. A second challenge was selecting variables with different reference years and definitions. A third involved interpreting workforce measures that may reflect regional referral centers rather than care available only to county residents. The analysis addresses these issues through explicit exclusions, a data dictionary, standardized rates, reproducible code, and cautious language.

Future Uses and Additional Applications

A future version could add preterm birth, cesarean delivery, severe maternal morbidity proxies, maternity care target areas, travel-time measures, Medicaid coverage, and state policy context. A yearly county panel could test whether changes in access precede changes in outcomes. Spatial validation could assess regional clustering. Local health systems could combine the screening model with service utilization and community needs-assessment data after governance and privacy review.

Recommendations

- Use the model as a screening layer that flags counties for human review, not as an automatic funding rule.
- Prioritize counties where elevated low birth weight overlaps with delayed prenatal care, poverty, or limited obstetric capacity.
- Validate flagged counties with local stakeholders, current service availability, patient travel patterns, and qualitative evidence.
- Track a small set of transparent indicators over time and publish definitions, denominators, exclusions, and data dates with every update.
- Re-estimate and recalibrate the model when new natality and AHRF releases become available.

Implementation Plan

Phase	Work	Output	Timing
1	Refresh and validate source extracts	Versioned county file and quality log	Weeks 1-2
2	Review indicators with public health and clinical partners	Approved definitions and decision criteria	Weeks 3-4
3	Pilot county screening and local validation	Shortlist with documented context	Weeks 5-8
4	Select outreach or capacity actions	Action plan with owners and measures	Weeks 9-12
5	Monitor results and audit equity impacts	Quarterly review and annual model refresh	Ongoing

Ethical Assessment

This analysis uses de-identified public aggregates, but ethical risk remains. County rankings can stigmatize communities, obscure structural causes, and encourage decision makers to treat model output as fact. Aggregation also masks within-county differences by race, income, disability, language, and neighborhood. Resource allocation based only on model scores could reproduce existing inequities because smaller rural counties are not individually identified in CDC WONDER.

Risk controls should include human review, public documentation, subgroup assessment where disclosure rules permit, community participation, and an appeal or correction process for local agencies. Reports should emphasize systems and access conditions, avoid blaming patients, suppress restricted counts, and state that predictive importance is not causal evidence. Model outputs should not be used to determine individual eligibility, clinical treatment, insurance coverage, or punitive oversight.

Appendix A: Supporting Documentation

Selected Data Dictionary

Variable	Definition	Source
fips	5-character county FIPS code	CDC WONDER / HRSA AHRF
low_birth_weight_pct	Births below 2,500 grams as a percent of births with known weight, pooled 2020-2024	CDC WONDER
late_or_no_prenatal_pct	Prenatal care beginning after month 3 or no prenatal care, percent of known records	CDC WONDER
tobacco_use_pct	Maternal tobacco use during pregnancy, percent of known records	CDC WONDER
maternal_age_15_19_pct	Births to mothers age 15-19, percent of age-classified births	CDC WONDER
maternal_age_35_plus_pct	Births to mothers age 35-54, percent of age-classified births	CDC WONDER
poverty_pct	Persons below the federal poverty level, ACS 2019-2023 (%)	HRSA AHRF
uninsured_under65_pct	Persons under 65 without health insurance, 2022 (%)	HRSA AHRF
obgyn_per_100k	Nonfederal general OB-GYN physicians in patient care per 100,000 residents, 2023	HRSA AHRF
midwives_per_100k	Advanced practice nurse midwives with an NPI per 100,000 residents, 2024	HRSA AHRF
obstetric_hospitals_per_100k	Short-term general hospitals with obstetric care per 100,000 residents, 2023	HRSA AHRF
log_population_density	Natural log of 1 plus population per square mile, 2020	HRSA AHRF
primary_care_hpsa	Indicator that all or part of the county had a primary care HPSA code in 2025	HRSA AHRF

Ten Questions an Audience May Ask

1. Why was low birth weight selected as the primary outcome?

2. How does CDC suppression affect the county estimates?

3. Why were the years 2020-2024 pooled?

4. Why are many rural counties absent from the model?

5. What does an out-of-sample R-squared of 0.57 mean for practical use?

6. Why did the random forest outperform ridge regression?

7. Does high feature importance mean a factor causes low birth weight?

8. How should a health department validate a flagged county?

9. What additional data would improve the model?

10. How will the project prevent county rankings from stigmatizing communities?

References

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